

Nonparametric estimation of quantiles of the conditional residual lifetime distribution

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- ▶ In medical research, one is often interested in studying **association between event times T_1 and T_2** , potentially subject to (right-)censoring
- ▶ Strength of association can be expressed in terms of **global and local association measures**, e.g.,
 - ▶ Kendall's tau
 - ▶ Cross-ratio function (Clayton, 1978)
- ▶ Alternatively, one can study **(quantiles of) the conditional residual lifetime distribution** (e.g., Abrams et al., 2021):

Examples

What is the median residual time to the occurrence of an event (i.e., represented by T_1), given that $T_2 \in (t_{21}, t_{22}]$:

- ▶ Time to breakthrough infection, conditional on humoral immunity level
- ▶ Time to blindness of one eye, conditional on time to blindness in the other one

Notation:

- ▶ Let (T_1, T_2) represent a random vector with $T_1 \geq 0$, $T_2 \geq 0$ and
 - ▶ $F(t_1, t_2) = P(T_1 \leq t_1, T_2 \leq t_2)$ the **joint cumulative distribution function**;
 - ▶ $F_1(t_1) = P(T_1 \leq t_1)$ and $F_2(t_2) = P(T_2 \leq t_2)$ **marginal distribution functions**
- ▶ Conditional distribution of T_1 given that $T_2 \leq t_2$:

$$F(t_1 \mid T_2 \leq t_2) = P(T_1 \leq t_1 \mid T_2 \leq t_2) = \frac{F(t_1, t_2)}{F_2(t_2)} =: \tilde{F}_{t_2}(t_1)$$

- ▶ **Conditional residual lifetime distribution** of T_1 at t_1 , given that $T_2 \leq t_2$:

$$P(T_1 - t_1 \leq y \mid T_1 > t_1, T_2 \leq t_2) = \frac{\tilde{F}_{t_2}(t_1 + y) - \tilde{F}_{t_2}(t_1)}{1 - \tilde{F}_{t_2}(t_1)}$$

- ▶ For $0 < p < 1$, we define the p -th quantile of the conditional residual lifetime of T_1 , given that $T_2 \leq t_2$ as:

$$\begin{aligned} Q(p | t_1, t_2) &= \inf \left\{ y : \frac{\tilde{F}_{t_2}(t_1 + y) - \tilde{F}_{t_2}(t_1)}{1 - \tilde{F}_{t_2}(t_1)} \geq p \right\} \\ &= -t_1 + \tilde{F}_{t_2}^{-1} \left[p + (1 - p)\tilde{F}_{t_2}(t_1) \right] \end{aligned}$$

- ▶ Nonparametric estimation of $Q(p | t_1, t_2)$ requires **nonparametric estimators for the conditional distribution function** $\tilde{F}_{t_2}$ and its quantile function (relying on the Akritas-Van Keilegom estimator for $F(t_1, t_2)$; see, e.g., Abrams et al., 2021)
- ▶ Abrams et al. (2021): estimators under **one-component censoring in T_1** only; asymptotic normality of the estimator

- ▶ An **extension of the work by Abrams et al. (2021)** to account for
 - ▶ Univariate censoring next to one-component censoring
 - ▶ Conditioning on T_2 of the form $\{t_{21} < T_2 \leq t_{22}\}$ instead of $\{T_2 \leq t_2\}$ (or $T_2 > t_2$)
 - ▶ Direct estimators for the joint distribution function $F(t_1, t_2)$ instead of the Akritas-Van Keilegom estimator
- ▶ We propose:
 - ▶ Two estimators for bivariate time-to-event data subject to **one-component censoring**
 - ▶ Two estimators for bivariate time-to-event data subject to **univariate censoring**

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- ▶ Danish twin registry data including **7955 monozygotic (MZ) and dizygotic (DZ) twin pairs** born between January 1, 1870 and December 31, 1930

Sex	Zygoty		Total
	MZ	DZ	
Male	1344	2411	3755
Female	1470	2730	4200
Total	2814	5141	7955

- ▶ **Sex, zygoty, and cause of death** (if applicable) reported
 - ▶ T_1 : time to death for twin member one
 - ▶ T_2 : time to death for twin member two
 - ▶ C : **univariate right censoring** of members of the twin pairs
- ▶ **Goal**: Estimate quantiles of the residual survival time for twin member, conditional on survival time for other member (based on T_2)

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- Conditional distribution of T_1 given that $t_{21} < T_2 \leq t_{22}$:

$$\begin{aligned}\tilde{F}(t_1 \mid t_{21}, t_{22}) &:= P(T_1 \leq t_1 \mid t_{21} < T_2 \leq t_{22}) \\ &= \frac{F(t_1, t_{22}) - F(t_1, t_{21})}{F_2(t_{22}) - F_2(t_{21})}\end{aligned}\tag{1}$$

- **Conditional residual lifetime distribution** of T_1 at t_1 , given that $t_{21} < T_2 \leq t_{22}$:

$$\begin{aligned}F(y \mid t_1, t_{21}, t_{22}) &= P(T_1 - t_1 \leq y \mid T_1 > t_1, t_{21} < T_2 \leq t_{22}) \\ &= \frac{\tilde{F}(t_1 + y \mid t_{21}, t_{22}) - \tilde{F}(t_1 \mid t_{21}, t_{22})}{1 - \tilde{F}(t_1 \mid t_{21}, t_{22})}\end{aligned}$$

- For $0 < p < 1$, we define the p -th quantile of the conditional residual lifetime of T_1 , given that $t_{21} < T_2 \leq t_{22}$ as:

$$\begin{aligned} Q(p \mid t_1, t_{21}, t_{22}) &= \inf \{y : F(y \mid t_1, t_{21}, t_{22}) \geq p\} \\ &= -t_1 + \tilde{F}^{-1} \left[p + (1 - p) \tilde{F}(t_1 \mid t_{21}, t_{22}) \mid t_{21}, t_{22} \right], \end{aligned} \quad (2)$$

with $\tilde{F}^{-1}$ the inverse of $\tilde{F}$.

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- ▶ Our estimators for $F(t_1, t_2)$ under **one-component and univariate censoring** will follow as particular cases of two more general estimators for $F(t_1, t_2)$
- ▶ (T_1, T_2) is subject to **independent censoring** by a vector (C_1, C_2)
- ▶ The **observable variables** are $(Z_1, Z_2, \Delta_1, \Delta_2)$ where
 - ▶ Observation times $Z_1 = T_1 \wedge C_1$ and $Z_2 = T_2 \wedge C_2$
 - ▶ Censoring indicators $\Delta_1 = I(T_1 \leq C_1)$ and $\Delta_2 = I(T_2 \leq C_2)$

① Inverse-probability weighted (IPW) estimator:

- ▶ The estimator of Burke (1988) is derived from the theoretical relation

$$F(t_1, t_2) = \int_0^{t_1} \int_0^{t_2} \frac{dP(T_1 \leq y_1, T_2 \leq y_2, \Delta_1 = 1, \Delta_2 = 1)}{S_G(y_1-, y_2-)}$$

- ▶ The **corresponding estimator** for a sample $(Z_{1i}, Z_{2i}, \Delta_{1i}, \Delta_{2i})$, $i = 1, \dots, n$, is

$$\hat{F}_B(t_1, t_2) = \frac{1}{n} \sum_{i=1}^n \frac{\Delta_{1i} \Delta_{2i}}{\hat{S}_G(Z_{1i-}, Z_{2i-})} I(Z_{1i} \leq t_1, Z_{2i} \leq t_2),$$

where $\hat{S}_G(t_1, t_2)$ is a nonparametric estimator for $S_G(t_1, t_2) = P(C_1 > t_1, C_2 > t_2)$

① Inverse-probability weighted (IPW) estimator:

- Under **one-component censoring** with $Z_2 = T_2$ and $\Delta_2 = 1$, we have

$$\hat{F}_{B1}(t_1, t_2) = \frac{1}{n} \sum_{i=1}^n \frac{\Delta_{1i}}{1 - \hat{G}(Z_{1i}-)} I(Z_{1i} \leq t_1, T_{2i} \leq t_2),$$

where $\hat{G}$ the Kaplan-Meier estimator for $G(c) = P(C_1 \leq c)$ based on the observations $(Z_{1i}, 1 - \Delta_{1i})$ ($i = 1, \dots, n$)

- Under **univariate censoring** with $C_1 = C_2 = C$ and $S_G(t_1, t_2) = 1 - G[(t_1 \vee t_2)-]$, we have

$$\hat{F}_{B2}(t_1, t_2) = \frac{1}{n} \sum_{i=1}^n \frac{\Delta_{1i} \Delta_{2i}}{1 - \hat{G}[(Z_{1i} \vee Z_{2i})-]} I(Z_{1i} \leq t_1, Z_{2i} \leq t_2),$$

where $\hat{G}$ is the Kaplan-Meier estimator of G based on the observations $((T_{1i} \vee T_{2i}) \wedge C_i, \Delta_i = 1 - \Delta_{1i} \Delta_{2i}) = (Z_{1i} \vee Z_{2i}, \Delta_i)$ ($i = 1, \dots, n$)

② Lin-Ying estimator:

- The estimator of Lin and Ying is derived from the following theoretical relation

$$P(Z_1 > t_1, Z_2 > t_2) = S(t_1, t_2)S_G(t_1, t_2),$$

where $S(t_1, t_2) = P(T_1 > t_1, T_2 > t_2)$ is the joint survival function of (T_1, T_2)

- Using this relation, $S(t_1, t_2)$ is estimated as

$$\hat{S}_{LY}(t_1, t_2) = \frac{1}{n} \frac{1}{\hat{S}_G(t_1, t_2)} \sum_{i=1}^n I(Z_{1i} > t_1, Z_{2i} > t_2),$$

where $\hat{S}_G(t_1, t_2)$ is a suitable nonparametric estimator of $S_G(t_1, t_2)$.

- This leads to an estimator for $F(t_1, t_2)$ using

$$\hat{F}_{LY}(t_1, t_2) = \hat{S}_{LY}(t_1, t_2) - \hat{S}_{LY}(t_1, 0) - \hat{S}_{LY}(0, t_2) + 1$$

Estimators of the joint distribution function $F(t_1, t_2)$

② Lin-Ying estimator:

- For **one-component censoring**, we have:

$$\begin{aligned}\hat{F}_{LY1}(t_1, t_2) = & \left\{ \frac{1}{n} \sum_{i=1}^n I(T_{2i} \leq t_2) \right\} \\ & - \left\{ \frac{1}{n} \frac{1}{[1 - \hat{G}(t_1)]} \sum_{i=1}^n I(Z_{1i} > t_1, T_{2i} \leq t_2) \right\}\end{aligned}$$

- For **univariate censoring**, we have:

$$\begin{aligned}\hat{F}_{LY2}(t_1, t_2) = & \left\{ \frac{1}{n} \frac{1}{[1 - \hat{G}(t_1 \vee t_2)]} \sum_{i=1}^n I(Z_{1i} > t_1, Z_{2i} > t_2) \right\} \\ & - \left\{ \frac{1}{n} \frac{1}{[1 - \hat{G}(t_1)]} \sum_{i=1}^n I(Z_{1i} > t_1) \right\} \\ & - \left\{ \frac{1}{n} \frac{1}{[1 - \hat{G}(t_2)]} \sum_{i=1}^n I(Z_{2i} > t_2) \right\} + 1\end{aligned}$$

- ▶ For the **numerator in equation (1)** we use $\widehat{F}(t_1, t_{22}) - \widehat{F}(t_1, t_{21})$
- ▶ The **denomintator in equation (1)** is estimated by the difference of the Kaplan-Meier estimators for $F_2(t_2)$, evaluated at t_{22} and t_{21} , respectively
- ▶ Combining the four estimators for the numerator with the appropriate estimator for the denominator leads to estimators $\widetilde{F}_n$ for $\widetilde{F}$.
- ▶ Finally, plugging in each of these estimators $\widetilde{F}_n$ (and $\widetilde{F}_n^{-1}$ in equation (2)), leads to four estimators $Q_n(p \mid t_1, t_{21}, t_{22})$ for the quantile function $Q(p \mid t_1, t_{21}, t_{22})$:

$$Q_n(p \mid t_1, t_{21}, t_{22}) = -t_1 + \widetilde{F}_n^{-1}(p + (1 - p)\widetilde{F}_n(t_1 \mid t_{21}, t_{22}) \mid t_{21}, t_{22}).$$

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- For T_j , $j = 1, 2$, we use a **Weibull distribution** with $s_j = 1.5$ and $d_j = 0.5$ such that

$$F_j(t_j) = 1 - \exp(-d_j t_j^{s_j}), \quad j = 1, 2$$

- For $C = C_1$ (one-component censoring) or $C = C_1 = C_2$ (univariate censoring) we use a Weibull distribution with parameters $s_C = 1.5$ and $d_C = 0.15$ (**moderate censoring**)
- To arrive at **bivariate Weibull distributions**, we use the **Clayton copula**

$$C_\theta(u, v) = (u^{-\theta} + v^{-\theta} - 1)^{-1/\theta}, \quad \theta \geq 0,$$

such that $F(t_1, t_2) = C_\theta[F_1(t_1), F_2(t_2)]$

- We generate **$M = 500$ datasets** of **$n = 250$ observations** $(z_{1i}, z_{2i}, \delta_{1i}, \delta_{2i})$, $i = 1, \dots, n$ using the Clayton copula with θ corresponding to Kendall's τ equal to 0.2 ($\theta = 0.5$)
- **Other scenarios** (heavy censoring, different copulae, etc.) not shown

Simulation results – One-component censoring

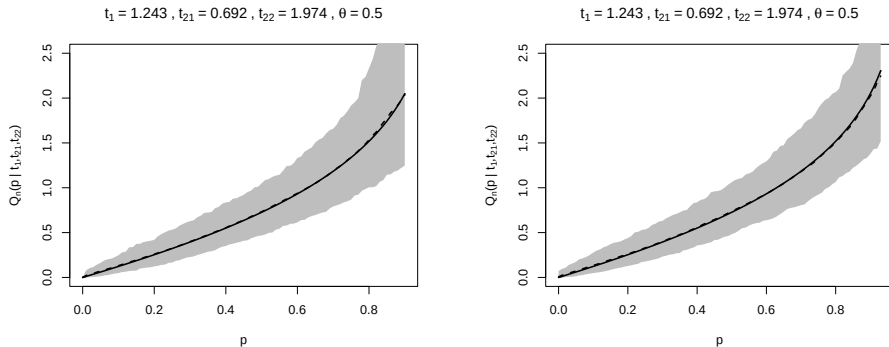


Figure: True quantile function $Q(p | t_1, t_{21}, t_{22})$ (black solid line) and averaged estimated quantile function $Q_n(p | t_1, t_{21}, t_{22})$ as a function of p (black dashed lines) for $t_1 = d_1^{-1/s_1} [\ln(2)]^{1/s_1} \approx 1.243$, $t_{21} \approx 0.692$ and $t_{22} \approx 1.974$ (moderate one-component censoring).

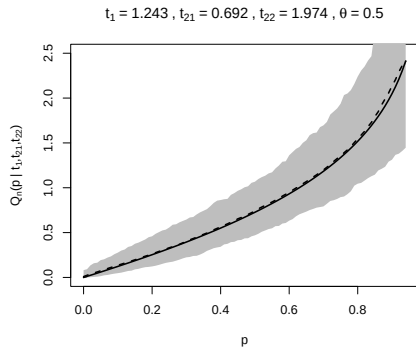
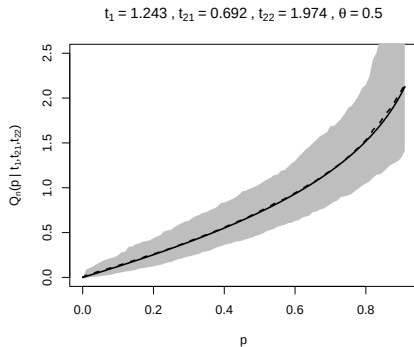
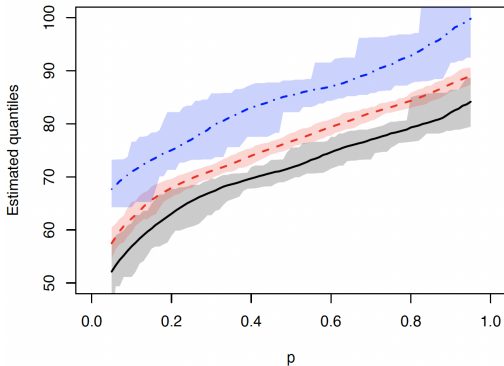


Figure: True quantile function $Q(p | t_1, t_{21}, t_{22})$ (black solid line) and averaged estimated quantile function $Q_n(p | t_1, t_{21}, t_{22})$ as a function of p (black dashed lines) for $t_1 = d_1^{-1/s_1} [\ln(2)]^{1/s_1} \approx 1.243$, $t_{21} \approx 0.692$ and $t_{22} \approx 1.974$ (moderate univariate censoring).

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Data application - MZ and DZ males

$Q_{n,1}(p \mid 0, t_{21}, t_{22})$ – MZ (males)



$Q_{n,1}(p \mid 0, t_{21}, t_{22})$ – DZ (males)

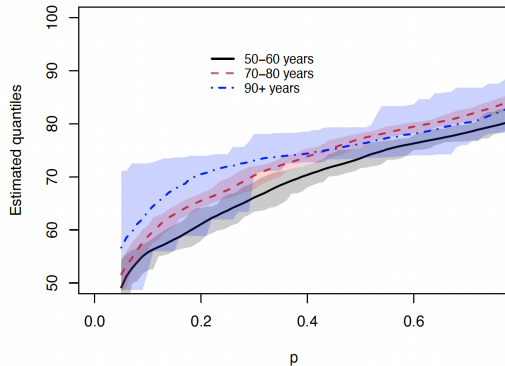
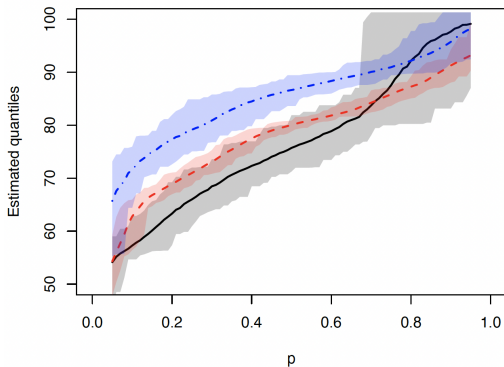


Figure: Estimated quantiles of the survival time ($t_1 = 0$) for MZ (left) and DZ (right) males.

$Q_{n,1}(p \mid 0, t_{21}, t_{22})$ – MZ (females)



$Q_{n,1}(p \mid 0, t_{21}, t_{22})$ – DZ (females)

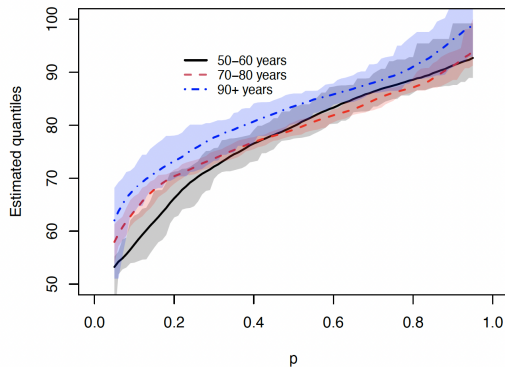
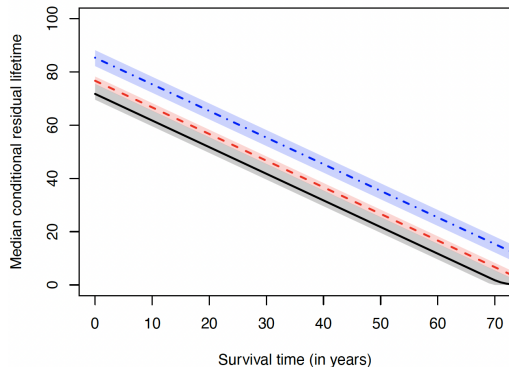


Figure: Estimated quantiles of the survival time ($t_1 = 0$) for MZ (left) and DZ (right) females.

Data application - MZ and DZ males

$Q_{n,1}(0.5 | 0, t_{21}, t_{22}) - \text{MZ (males)}$



$Q_{n,1}(0.5 | 0, t_{21}, t_{22}) - \text{MZ (males)}$

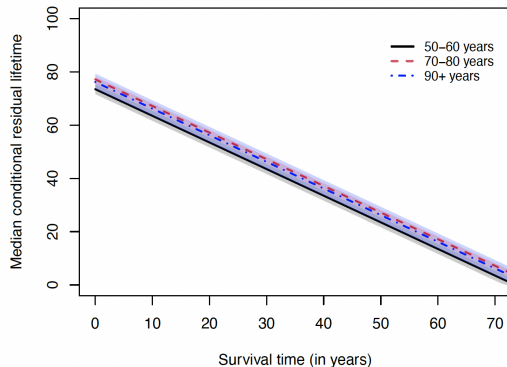


Figure: Estimated median conditional residual survival time for MZ (left) and DZ (right) females.

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- ▶ Good **finite-sample performance** of proposed estimators
- ▶ Estimators are able to quantify differences in (quantiles of) the conditional residual lifetime distribution across relevant groups, which is of crucial importance in, for example, **prognostic research**
- ▶ Instead of relying on the **difference of the Kaplan-Meier estimator** for $F_2(t_2)$, evaluated in t_{22} and t_{21} , respectively, to estimate the denominator in equation (1), one could use $\hat{F}(\infty, t_{22}) - \hat{F}(\infty, t_{21})$ as alternative estimator
- ▶ In contrast to the Kaplan-Meier based estimator(s), the estimator itself is **not a well-defined cumulative distribution function** with values approaching one for $t_1 \rightarrow \infty$
- ▶ Although computationally plausible, and providing consistent estimation of the quantiles of the conditional residual lifetime distribution (not shown here), we did not consider such estimators given that their **finite-sample performance was suboptimal** as compared to that of the Kaplan-Meier based estimators

- ▶ Estimators of the quantiles of the conditional residual lifetime distribution can be derived in the **presence of (double) truncation**
- ▶ More specifically, estimation of the conditional distribution function of T_1 , given that $t_{21} < T_2 \leq t_{22}$ could, for example, be based on the **nonparametric maximum likelihood estimator proposed by Efron and Petrosian (1999)**
- ▶ An extension to encompass **random bivariate censoring based on (C_1, C_2)** is an interesting topic for further research
- ▶ Finally, investigating the use of these techniques in the context of **surrogate marker evaluation for survival endpoints** (semi-competing risks)

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